

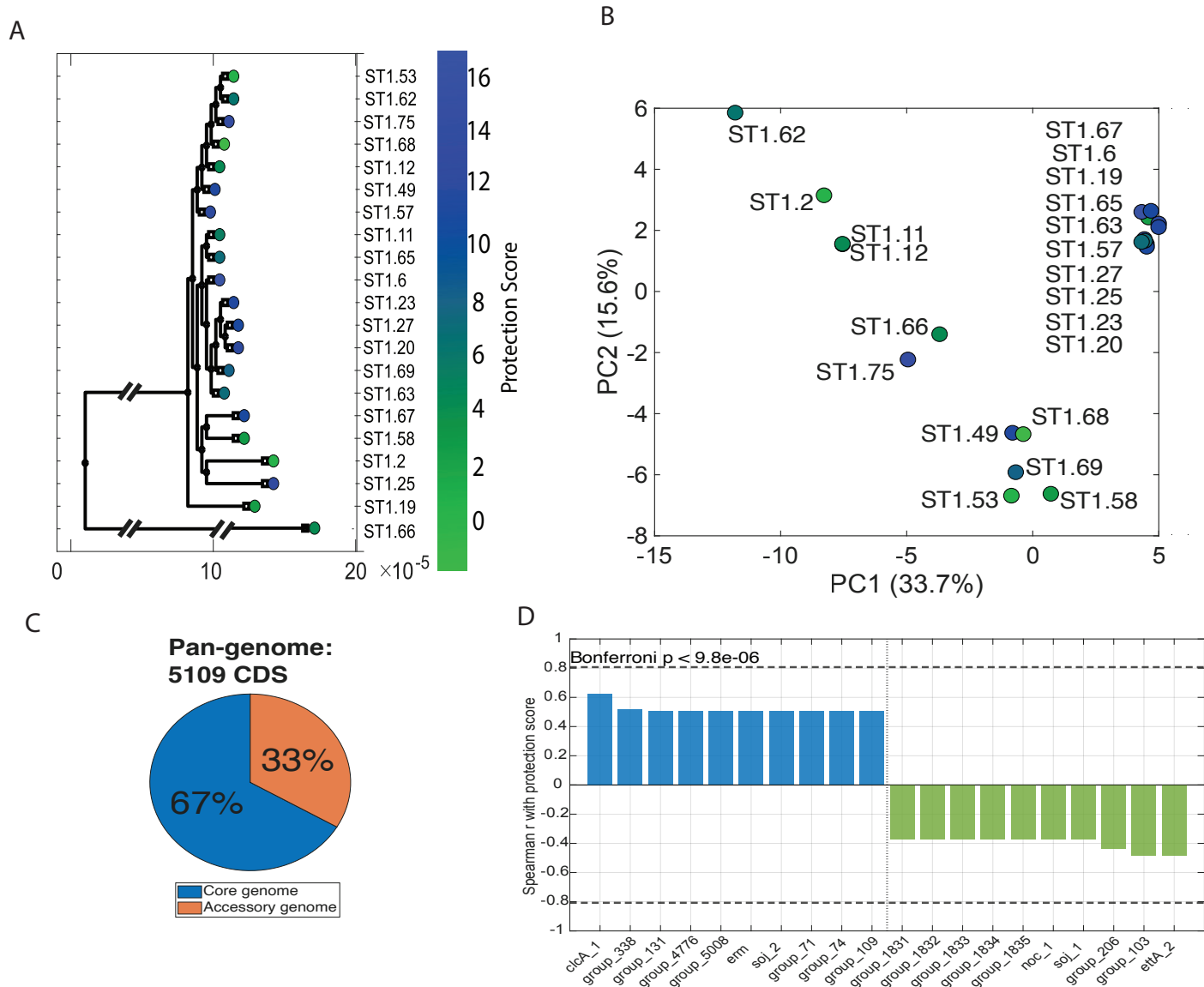


Figure 4: Genomic analysis of ST1 strains does not reveal a single determinant of protection. (A) Phylogenetic tree of 21 ST1 clinical isolates based on core genome single-nucleotide substitutions. Branch tips are colored by protection score (color scale, right). No distinct phylogenetic clustering separates protective from non-protective strains. (B) Principal component analysis (PCA) of the accessory genome presence/absence matrix for all 21 ST1 strains. PC1 explains 33.7% and PC2 explains 15.6% of the variance. Strains are colored by protection score as in (A). No clear separation between protective and non-protective strains is observed. (C) Pan-genome composition of the 21 ST1 strains, comprising 5,109 total coding sequences (CDS). The core genome (shared across all strains, blue) accounts for 67% of CDS, and the accessory genome (present in a subset of strains, orange) accounts for 33%. (D) Spearman correlation coefficients between individual gene presence/absence and protection score across all 21 ST1 strains. Blue bars indicate positively correlated genes (present in more protective strains) and green bars indicate negatively correlated genes. Dashed lines indicate the Bonferroni-corrected significance threshold ($p < 9.8 \times 10^{-6}$). Genes shown are those passing the significance threshold ($p < 0.05$).